ASSIGNMENT-20.2

NAME: HABEEBA KHANAM

HALL TICKET :2303A51474

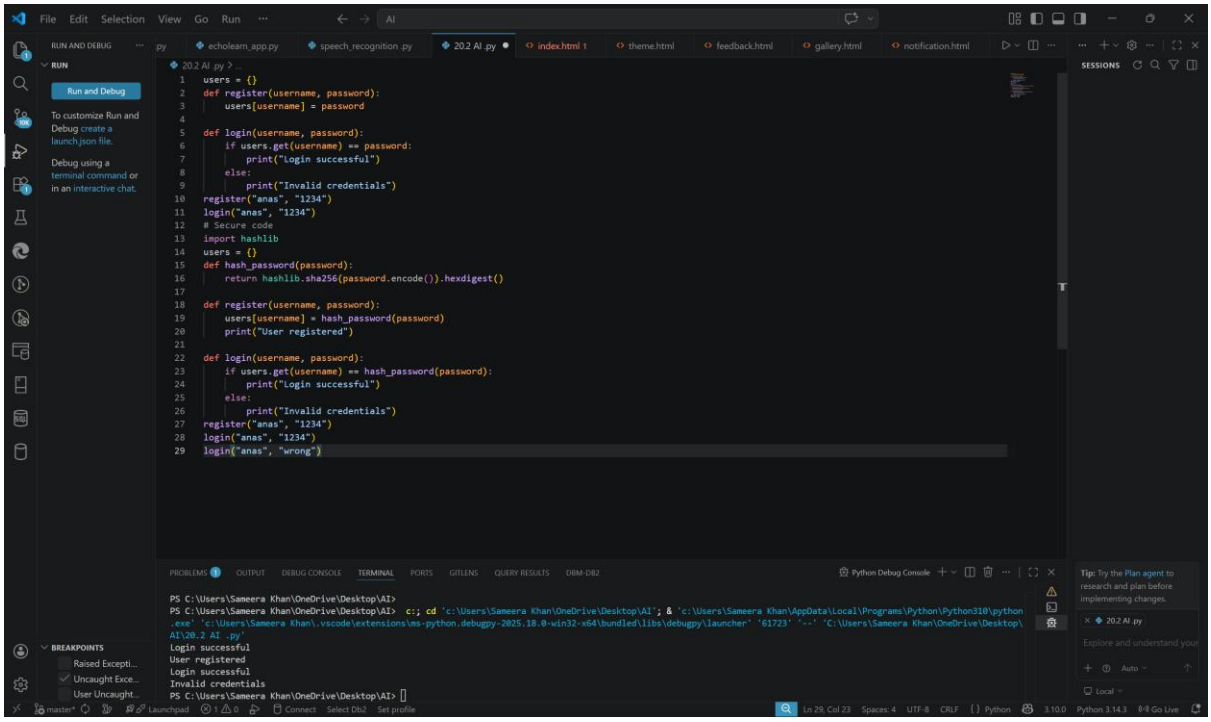
BATCH NO:29

LAB 20 – SECURITY TESTING: IDENTIFYING VULNERABILITIES IN AI-

TASK 1 – PASSWORD SECURITY ENHANCEMENT

GENERATED CODE

CODE:

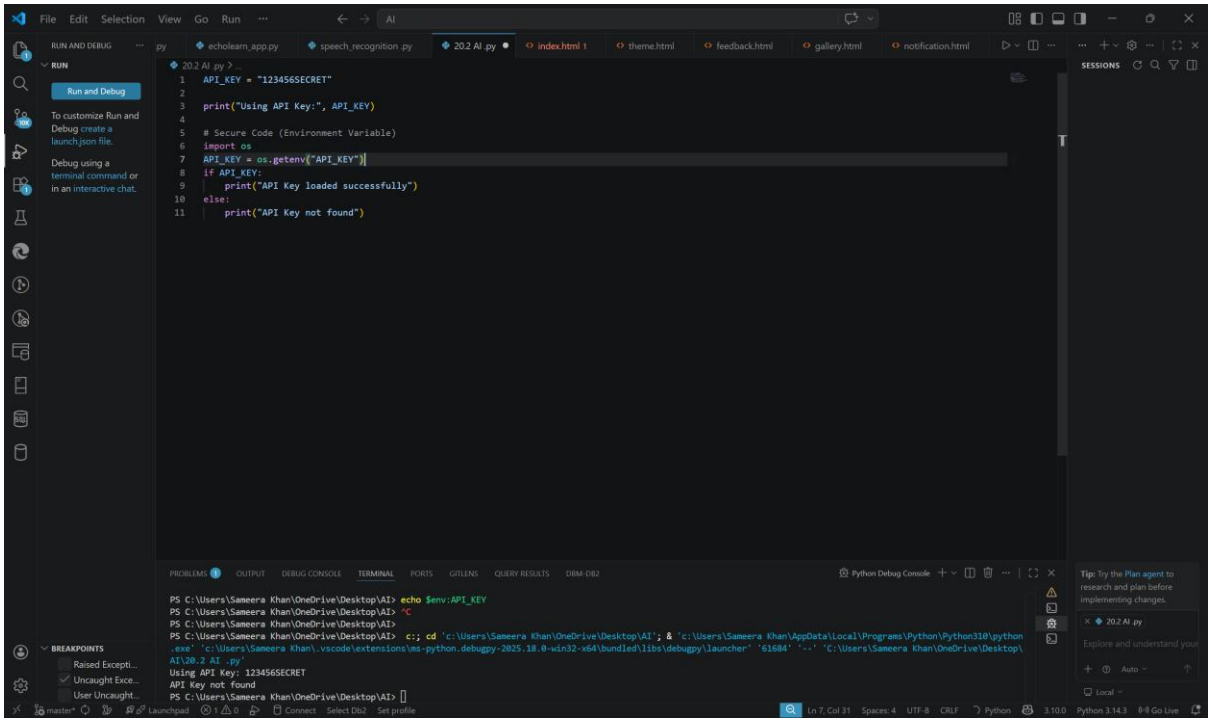


OBSERVATION:

Passwords stored in plain text are vulnerable to theft if the database is compromised. Using hashing (SHA-256) secures passwords by storing irreversible encrypted values instead of actual passwords.

TASK 2 – SECURE API KEY MANAGEMENT

Code:

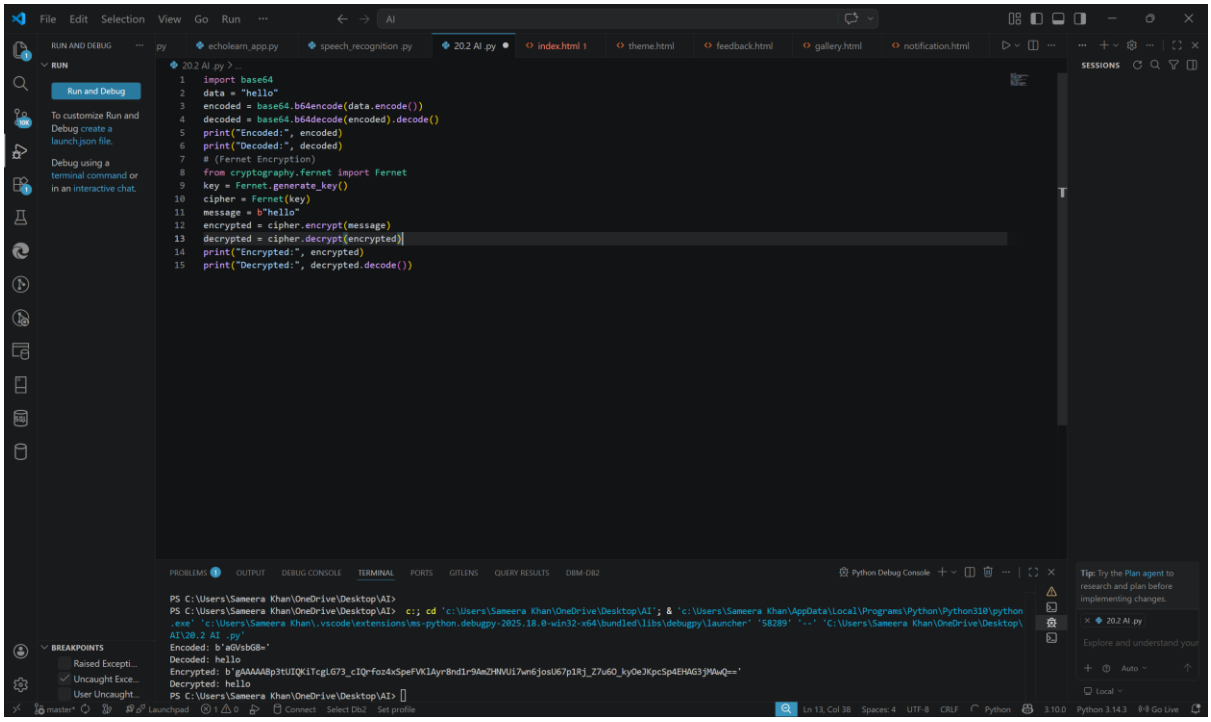


Observation:

Hardcoding API keys in source code exposes sensitive credentials to anyone with access to the code.
Using environment variables keeps API keys secure and prevents accidental leakage.

TASK 3 – WEAK ENCRYPTION DETECTION

CODE:

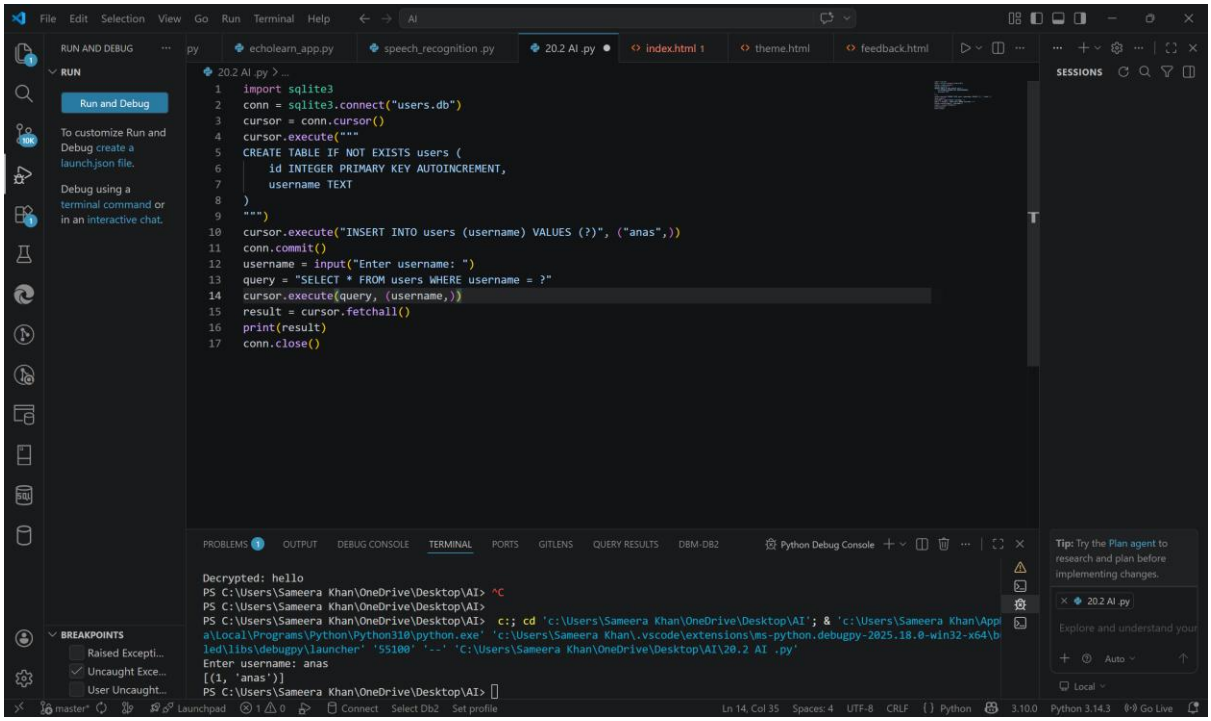


OBSERVATION:

Basic encoding methods like Base64 are not secure as they can be easily reversed.
Strong encryption (Fernet) ensures data confidentiality and requires a key for decryption.

TASK 4 – REAL-TIME APPLICATION: SECURITY REVIEW OF AI-GENERATED WEB SERVICE

CODE:

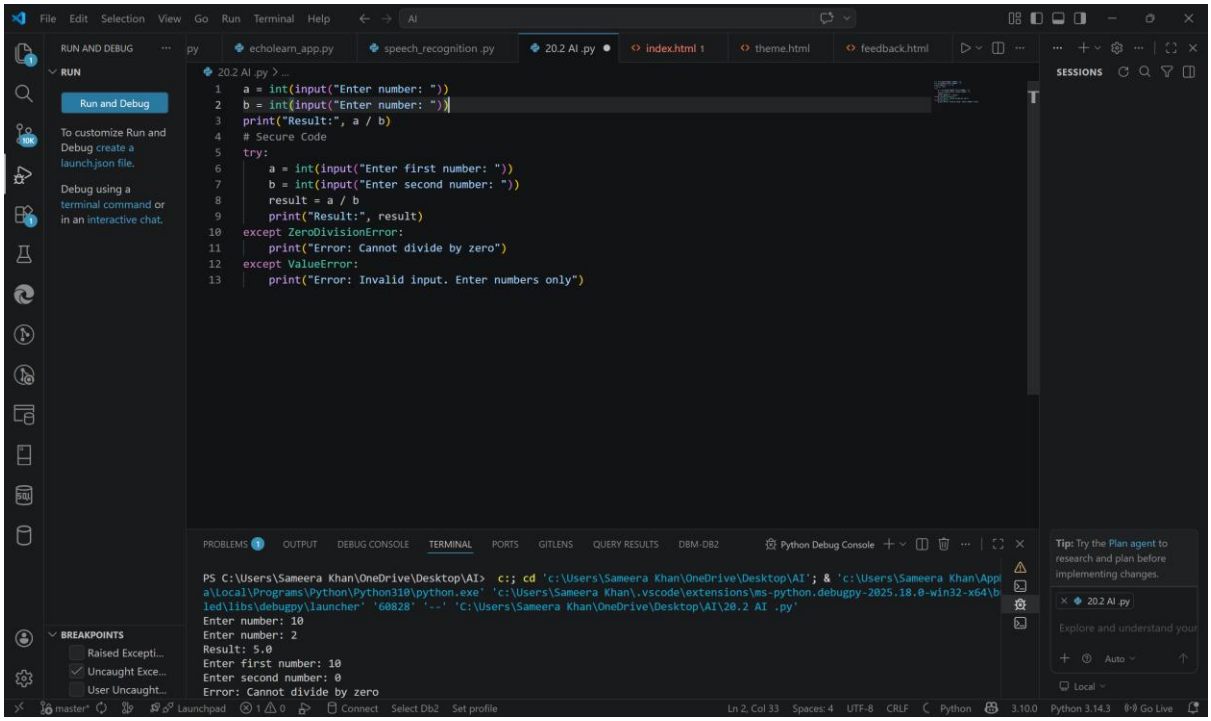


OBSERVATION:

Building SQL queries using user input directly leads to SQL injection vulnerabilities. Using parameterized queries and validation prevents unauthorized database access.

TASK 5 – EXCEPTION HANDLING IMPROVEMENT

CODE:



OBSERVATION:

Programs without exception handling may crash on invalid input or runtime errors.
Using try-except blocks improves reliability and provides user-friendly error messages.